

The empirical recurrence for OEIS A251339: recovering a conjecture that is not written down

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Abstract

OEIS A251339 records “Empirical recurrence of order 41”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 1024$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 41, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 41 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A251339 is “Number of $(n+1) \times (4+1)$ 0..3 arrays with every 2×2 subblock summing to a nonzero multiple of 3.”. It has offset 1 and begins

13275, 189129, 3014066, 55048995, 1151715397, 27005232676, 687109914883, 18427859055437, ...

The entry states:

Empirical recurrence of order 41 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 10:22:08, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1024$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 1024$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 2048$ exact terms of the model returns order 41 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{41} c_i a(n-i),$$

c_1	180	c_{12}	−2229608267330935	c_{23}	178686912801152157340	c_{34}	−470405629896
c_2	−14497	c_{13}	1930357663245910	c_{24}	−29762441194938239301572	c_{35}	5055049018299
c_3	684788	c_{14}	80951363027209241	c_{25}	−28412013877486219846580	c_{36}	1819379399673
c_4	−20778546	c_{15}	−256389871651176138	c_{26}	67960714947968681926980	c_{37}	−121552842683
c_5	411788044	c_{16}	−1862598964566334054	c_{27}	115315136686172023312304	c_{38}	−328902129068
c_6	−4979718142	c_{17}	7899399023021815384	c_{28}	−65317355279979474457968	c_{39}	1689663026534
c_7	24098796170	c_{18}	34199260129607471306	c_{29}	−211005568360031270775504	c_{40}	208437615216
c_8	258269792895	c_{19}	−129646464395489786704	c_{30}	−270250090290856153056	c_{41}	−94961113576
c_9	−5102299252812	c_{20}	−539314105531791599035	c_{31}	214915586323738114831104		
c_{10}	24163533625143	c_{21}	1023121349054226763202	c_{32}	53881962059058016646016		
c_{11}	164952596096496	c_{22}	5967704821238071462945	c_{33}	−131856473977413921944064		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 41, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $41 + 4$ terms removed returns order 41 as well, so $\deg q_{\min} = 41 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $41 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 41 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A251339.

- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.